

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                      |                          |
|----------------------|--------------------------|
| Product Name         | <u>Succinic acid</u>     |
| CAS No               | 110-15-6                 |
| Synonyms             | Butanedioic acid         |
| Molecular Formula    | C4 H6 O4                 |
| Molecular Weight     | 118.09                   |
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | 219550000; 219550010; 219552500                                                             |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                  |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

HSNO Approval Number     HSR002503

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

Serious Eye Damage/Eye Irritation

Category 1

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements

**Signal Word****Danger****Hazard Statements**

H318 - Causes serious eye damage

**Precautionary Statements****Prevention**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

## Section 3 - Composition and Information on Ingredients

| Component     | CAS No   | Weight % |
|---------------|----------|----------|
| Succinic acid | 110-15-6 | >95      |

## Section 4 - First Aid Measures

**Description of first aid measures****General Advice**

If symptoms persist, call a physician.

**New Zealand Emergency Tel.**CHEMTREC®  
09 980 6780 or +64 9 980 6780**Inhalation**

Remove to fresh air. If breathing is irregular or stopped, administer artificial respiration. Get medical attention.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Self-Protection of the First Aider**

Use personal protective equipment as required.

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|                                            |                                      |
|--------------------------------------------|--------------------------------------|
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom. |
| <b>Most important symptoms and effects</b> | Causes severe eye damage.            |
| <b>Notes to Physician</b>                  | Treat symptomatically.               |

## **Section 5 - Fire Fighting Measures**

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### **Decomposition Temperature**

> 235°C

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **Section 6 - Accidental Release Measures**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

#### **Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

#### **Methods for Containment and Clean Up**

Avoid dust formation. Sweep up and shovel into suitable containers for disposal.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

### **Precautions for Safe Handling**

#### **Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### **Conditions for Safe Storage, Including any Incompatibilities**

**Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Incompatible Materials**

Bases. Reducing Agent.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material                                               | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|--------------------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Butyl rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:  
Recommended half mask:-**

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)  
Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                         |                                                                                                                   |                                   |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Physical State                          | Powder Solid                                                                                                      |                                   |
| Appearance                              | White                                                                                                             |                                   |
| Odor                                    | Odorless                                                                                                          |                                   |
| Odor Threshold                          | No data available                                                                                                 |                                   |
| pH                                      | 2.7                                                                                                               | 0.1M aq.sol                       |
| Melting Point/Range                     | 185 - 190 °C / 365 - 374 °F                                                                                       |                                   |
| Softening Point                         | No data available                                                                                                 |                                   |
| Boiling Point/Range                     | 235 °C / 455 °F                                                                                                   |                                   |
| Flammability (liquid)                   | Not applicable                                                                                                    | Solid                             |
| Flammability (solid,gas)                | No information available                                                                                          |                                   |
| Explosion Limits                        | No data available                                                                                                 |                                   |
| Flash Point                             | 206 °C / 402.8 °F                                                                                                 | Method - No information available |
| Autoignition Temperature                | 630 °C / 1166 °F                                                                                                  |                                   |
| Decomposition Temperature               | > 235°C                                                                                                           |                                   |
| Viscosity                               | Not applicable                                                                                                    | Solid                             |
| Water Solubility                        | 80 g/L (20°C)                                                                                                     |                                   |
| Solubility in other solvents            | No information available                                                                                          |                                   |
| Partition Coefficient (n-octanol/water) |                                                                                                                   |                                   |
| Vapor Pressure                          | 0.0022 mmHg @ 19 °C                                                                                               |                                   |
| Density / Specific Gravity              | 1.56                                                                                                              |                                   |
| Bulk Density                            | No data available                                                                                                 |                                   |
| Vapor Density                           | Not applicable                                                                                                    | Solid                             |
| Particle characteristics                | No data available                                                                                                 |                                   |
| <b>Other information</b>                |                                                                                                                   |                                   |
| Molecular Formula                       | C4 H6 O4                                                                                                          |                                   |
| Molecular Weight                        | 118.09                                                                                                            |                                   |
| Explosive Properties                    | Not explosive Dust can form an explosive mixture with air                                                         |                                   |
| Oxidizing Properties                    | Not oxidising (based on the chemical structure of the substance and oxidation states of the constituent elements) |                                   |
| Evaporation Rate                        | Not applicable - Solid                                                                                            |                                   |

## Section 10 - Stability and Reactivity

|                                  |                                                           |
|----------------------------------|-----------------------------------------------------------|
| Reactivity                       | None known, based on information available                |
| Stability                        | Stable under normal conditions.                           |
| Sensitivity to Mechanical Impact | No information available                                  |
| Sensitivity to Static Discharge  | No information available                                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                  |
| Hazardous Reactions              | None under normal processing.                             |
| Conditions to Avoid              | Avoid dust formation, Incompatible products, Excess heat. |
| Incompatible Materials           | Bases, Reducing Agent.                                    |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).  |

## Section 11 - Toxicological Information

### Acute Effects

#### Information on likely routes of exposure

#### Product Information

|            |                                                                                                                                                                  |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation | Irritating to respiratory system. May be harmful if inhaled. May cause irritation of respiratory tract.                                                          |
| Eyes       | Irritating to eyes.                                                                                                                                              |
| Skin       | Irritating to skin. May be harmful in contact with skin.                                                                                                         |
| Ingestion  | May be harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Ingestion may cause irritation to mucous membranes. |

#### Numerical measures of toxicity

##### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Based on available data, the classification criteria are not met |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

| Component     | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation                       |
|---------------|---------------------------|---------------------------|---------------------------------------|
| Succinic acid | LD50 = 2260 mg/kg ( Rat ) | LD50 = 6740 mg/kg ( Rat ) | LC50 = 1284 mg/m <sup>3</sup> ( Rat ) |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

Not mutagenic in AMES Test

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

#### Symptoms / effects, both acute and delayed

No information available.

## Section 12 - Ecological Information

### Ecotoxicity

#### Aquatic ecotoxicity

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component     | Freshwater Fish                                | Water Flea                                       | Freshwater Algae                                                   | Microtox                                           |
|---------------|------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------|
| Succinic acid | LC50 > 100 mg/l 96h<br>(Danio rerio)(OECD 203) | EC50 > 100 mg/l 48h<br>(Daphnia magna)(OECD 202) | EC50 > 100 mg/l 72h<br>(Pseudokirchneriella subcapitata)(OECD 201) | EC50 > 300 mg/l 3h<br>(activated sludge)(OECD 209) |

#### Terrestrial ecotoxicity

There is no data for this product

#### Persistence and Degradability

Readily biodegradable

#### Persistence

Soluble in water, Persistence is unlikely, based on information available.

#### Bioaccumulative Potential

Bioaccumulation is unlikely

#### Mobility

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### Other adverse effects

#### Endocrine Disruptor Information Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

#### Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### Other Information

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer.

## Section 14 - Transport Information

#### NZS 5433:2020

Not regulated

#### IATA

Not regulated

|                                                                                 |                                                                                                                         |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IMDG/IMO</b>                                                                 | Not regulated                                                                                                           |
| <b>Environmental hazards</b>                                                    | No hazards identified                                                                                                   |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods                                                                                          |
| <b>Special Precautions</b>                                                      | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
| <b>Additional information</b>                                                   | None known                                                                                                              |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR002503 |
|-----------------------------|-----------|

#### **National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

### International Regulations

|                                                         |                                                                |
|---------------------------------------------------------|----------------------------------------------------------------|
| <b>Ozone Depletion Potential</b>                        | This product does not contain any known or suspected substance |
| <b>Persistent Organic Pollutant</b>                     | This product does not contain any known or suspected substance |
| <b>Rotterdam Convention (PIC)</b>                       | Not applicable                                                 |
| <b>Authorisation/Restrictions according to EU REACH</b> | Not applicable                                                 |

### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component     | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|---------------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| Succinic acid | 110-15-6 | X     | X    | 203-740-4 | -      | -   | KE-13150 | X     | X    |

| Component     | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|---------------|----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Succinic acid | 110-15-6 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |



Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date** 10-Mar-2023  
**Revision Summary** Not applicable

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet